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Murphy Express
(Alameda Blvd. / 2nd St.)

Traffic Study

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F I N A L

Presented to:

New Mexico Department of Transportation
District 3
&
Bernalillo County Public Works Dept.
&
City of Albuquerque Transportation Dev.

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Murphy Express (Alameda Blvd. / Second St.) Traffic Study

Executive Summary

The purpose of this study is to evaluate the transportation conditions after implementation of the proposed Murphy Express primarily to determine the feasibility of a single full access driveway on Alameda Blvd. at this location. This study is prepared to meet the requirements of the New Mexico Department of Transportation (NM DOT) and Bernalillo County Public Works Department associated with their review of the proposed Murphy Express facility.

The proposed development is located at the northwest corner of Alameda Blvd. / 2nd St. The study area includes the intersections of Alameda Blvd. / Fourth St. and Alameda Blvd. / Second St. as well as the proposed driveway for the project.

The proposed development is to be developed as a gasoline station with convenience market containing 12 fueling positions. The anticipated implementation year for this site is the year 2020. According to the Institute of Traffic Engineers' (ITE) trip generation rates, the weekday AM Peak Hour period is anticipated to generate 67 entering trips and 64 exiting trips during the AM Peak Hour. During the weekday PM Peak Hour period, it is anticipated that it will generate approximately 86 entering trips and 83 exiting trips. A 30% pass-by trip rate reduction was applied to those generated trips.

The development will be accessed via one proposed full-access driveway on Alameda Blvd. (Driveway "A"). In consideration of the heavy directional flow of traffic on Alameda Blvd. (predominantly eastbound during the AM Peak Hour period and predominantly westbound during the PM Peak Hour period), this Study will evaluate the operational characteristics of the proposed driveway for the project. The analysis of the driveway will consider the impact of the adjacent signalized intersections on the operation of the driveway (i.e., platooning effect from nearby signalized intersections). The adjacent intersections were analyzed using Multiple Period Analysis as required by the New Mexico Department of Transportation and the Highway Capacity Manual, 6th Edition.

Analysis results for the implementation year (worst period analysis) are summarized in the following table:

Intersection	2020 AM Peak Period		2020 PM Peak Period	
	NO BUILD	BUILD	NO BUILD	BUILD
1 – Alameda Blvd. / 4 th St.	C – 26.9	C – 30.6	D – 47.9	D – 50.5
2 – Alameda Blvd. / 2 nd St.	B – 13.1	B – 13.3	C – 28.0	C – 29.3
3 – Alameda Blvd. / Driveway “A”***	N/A	E – 46.2*	N/A	F – 503*
3a – Alameda Blvd. / Driveway “A”	N/A	A – 9.8	N/A	F – 135.4
4 – Driveway “B” / 2 nd St.***	N/A	A – 9.9	N/A	B – 10.1
4a – Driveway “B” / 2 nd St.	N/A	A – 9.6	N/A	A – 9.8

* - Level-of-Service / delay for unsignalized intersection is the highest delay movement (SB Left Turn Movement).

** - Results in Table above are based on a full access driveway on Alameda Blvd. for Murphy Express. The NM DOT has indicated that they will restrict access for Driveway “A” to right-in, right-out only. Analysis for the right-in, right-out only driveway are reported in 3a above.

*** - Results in Table above are based on a full access driveway on 2nd St. for Murphy Express. The Bernalillo County Public Works Department has indicated that they will restrict access for Driveway “B” to right-in, right-out only. Analysis for the right-in, right-out only driveway are reported in 4a above.

The worst periods for this corridor were Period 2 for the AM Peak (7:15 AM to 7:30 AM) and Period 5 for the PM Peak (5:00 PM – 5:15 PM).

In summary, the proposed unsignalized driveway onto Alameda Blvd. should be acceptable as a right-in, right-out, left-in only driveway. The HCM6 analysis of the proposed Driveway “A” onto Alameda Blvd. as a full access driveway shows that the operation of the driveway will be marginal during the AM Peak Period but will experience excessive delays during the PM Peak Period. The most severe movement is the southbound left turn movement exiting Driveway “A” desiring to travel eastbound on Alameda Blvd. Also, the eastbound left turn movement on Alameda Blvd. at proposed Driveway “A” shows to experience LOS “F” with an average control delay of 64.1 seconds during the PM Peak Period. However, a microsimulation of the Alameda corridor study area during the PM Peak Period confirms the excessive delays and queuing associated with the southbound to eastbound left turn movement at Driveway “A”, but reveals no major issue with the eastbound to northbound left turn movement off of Alameda Blvd. into Driveway “A”. The HCM calculated 95th Percentile Queue length in the report indicates a queue length of 1.1 vehicle (or 2 vehicles). The microsimulation shows a 95th Percentile Queue length of 2 vehicles as well. There appear to be substantial waiting times for the subject eastbound left movement, but those waiting times are limited by the influence of the upstream traffic signal at 2nd St. which create gaps in the westbound flow of traffic, thus permitted a couple of vehicles to negotiate the eastbound to northbound left turn movement at least once each signal cycle. This analysis finds that the eastbound to northbound left turn movement should be permissible at proposed Driveway “A”, but the southbound to eastbound left turn movement exiting Driveway “A” will experience excessive delays based on the calculations.

Alternate access east to Second Street is recommended. The problematic southbound left turn exiting traffic at Driveway “A” should have an alternative to exit the facility to drive east on Alameda Blvd., south on Second St., or north on Second St. The Second St. driveway (Driveway “B”) will provide remedy for the problematic exiting southbound to eastbound turning movement from Driveway “A”. The southbound to eastbound exiting left turn movement from Driveway “A” should be prohibited.

The New Mexico Department of Transportation District 3 Traffic Engineer has commented that Driveway “A” (on Alameda Blvd.) will be required to be restricted to a right-in, right-out only driveway. The Bernalillo County Public Works Department has commented that Driveway “B” (on Second St.) will be required to be restricted to a right-in, right-out only driveway.

In summary, the recommendations of this study are:

Recommendations:

Based on 2020 (Implementation Year) Analysis:

- An eastbound left turn deceleration lane and a westbound right turn deceleration lane are both warranted at the proposed Driveway “A” accessing the new Murphy Express facility. Based on the posted speed limit of 40 MPH on Alameda Blvd. along the frontage of this project, the New Mexico Department of Transportation’s State Access Management Manual stipulates that the right turn deceleration lane should be designed and constructed to a length of 300 feet including transition. It stipulates that the left turn deceleration lane should be designed and constructed to a length of 325 feet including transition. Field constraints imposed by urban conditions preclude construction of the right turn and left turn deceleration lanes warranted. This study recommends that the westbound right turn deceleration lane be designed and constructed to begin at the west curb return of the intersection of Alameda Blvd. / 2nd St. and extend about 200 feet to the new Murphy Express Driveway “A”. By virtue of the fact that the New Mexico Department of Transportation has prohibited the eastbound left turn movement into Driveway “A” off of Alameda Blvd., then warrant for the eastbound left turn lane is rendered meaningless.
- Provide a new connection to and from the proposed Murphy Express facility to a new driveway on Second St. which will be restricted to right-in, right-out only by the Bernalillo County Public Works Department. The new driveway on Second St. (Driveway “B”) should be located as far as possible north of Second St., but a minimum of 250 feet (centerline to centerline) if possible. Access from the Murphy Express facility to Second St. will require coordination and permitting from the Middle Rio Grande Conservancy District (MRCGD) to cross the existing drainage ditch along the west side of Second St.

**Murphy Express
(Alameda Blvd. / Second St.)
Traffic Study**

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Attachment: BA2026-0001_Additional Documents (26992 : ZBA2026-0001/ZA2026-0024)

Murphy Express (Alameda Blvd. / Second St.) Traffic Study

Introduction

The purpose of this study is to evaluate the transportation conditions after implementation of the proposed Murphy Express primarily to determine the feasibility of a single full access driveway on Alameda Blvd. at this location. This study is prepared to meet the requirements of the New Mexico Department of Transportation (NM DOT) and Bernalillo County Public Works Department associated with their review of the proposed Murphy Express facility. This study is prepared to meet the requirements of the NMDOT and Bernalillo County associated with their review of the Murphy Express facility as shown on the plan on Page A-3 in the Appendix of this report. A complimentary report will be submitted to the City of Albuquerque.

The proposed development is located at the northwest corner of Alameda Blvd. / Second St. in Albuquerque, New Mexico. If the property was to develop in a manner significantly different than the proposed plan considered in this report such that the number of generated trips is significantly greater, then an update to this study may be required by NMDOT or Bernalillo County.

Following is a vicinity map depicting the location of the proposed project:



Following is the proposed site development plan depicting driveway (access) location (also, see Appendix Page A-3 for a more complete version of the proposed site development plan):

Study Area Conditions

A Traffic Impact Study Scoping Meeting was held with the NMDOT (Nancy Perea and Margaret Haynes) and Bernalillo County PWD (Julie Luna). During the exchange, it was determined that the study area would include the following list of intersections to be analyzed in the Traffic Impact Study:

1. Alameda Blvd. / Fourth St.
2. Alameda Blvd. / Second St.
3. Alameda Blvd. / Driveway "A"

This scope of study was based on the assumption that the parcel in question would be developed as a 12 fueling position gasoline station with convenience market as shown on the proposed site plan. It was determined at the meeting by the New Mexico Department of Transportation that this would not be required to be a full Traffic Impact Study. Its purpose was primarily to determine the feasibility of allowing a full-access driveway for the proposed Murphy Express onto Alameda Blvd. In order to define the operation of the driveway, the impact of the adjacent traffic signals (i.e., platooning effect). Therefore, analysis of the signalized intersections of Alameda Blvd. / 4th St. and Alameda Blvd. / 2nd St. were necessary. Based on current New Mexico Department of Transportation requirements (and HCM6 requirements) for Traffic Studies of this nature, demand traffic count volumes were required as well as Multiple Period Analyses for the two signalized intersections. The results of this Study focused on the worst 15-minute periods (AM and PM) in the Multiple Period analyses.

There are no other known land development projects in the area which need to be incorporated into the background traffic model for this study. Also, there are no known Transportation Improvement Program projects in the area that need to be considered in the Traffic Impact Study.

This project is served by public transit services in the area; specifically Routes #98 and 10. These routes run along Alameda Blvd. (98-commuter) or Fourth St. (10).

Fourth St. & Second St. are designated on the Futures 2040 Metropolitan Transportation Plan (2040 Long Range Bikeway System) as Proposed Bicycle Lanes or Paved Trails. Alameda Blvd. in this area includes a bicycle lane. The Alameda Blvd. multi-use trail is on the south side of the roadway. See Appendix page A-5 for a portion of the map.

There are pedestrian facilities in the project area – curb & gutter and sidewalks along the roads, as well as raised medians for pedestrians & bicyclists crossing against traffic.

Second St. & Alameda Blvd. are classified as Regional Principal Arterial Roadways. Second St. north of Alameda Blvd. is a two-lane roadway with curb & gutter with sidewalks. Second St. south of Alameda is a four lane roadway with raised medians and left turn lanes at major intersections.

The posted speed limit along Second St. is 45 MPH. Alameda Blvd. is a four-lane roadway with raised medians and curb & gutter with sidewalks. The posted speed limit along Alameda Blvd. is 40 MPH.

Fourth St. is classified as a Minor Arterial Roadway on the Mid-Region Council of Government's Futures 2040 Long Range Roadway System Map. Fourth St. is generally a four-lane roadway with curb & gutter with sidewalks. The posted speed limit along this section of Fourth St. is 35 MPH.

Analysis of Existing Conditions

Due to the fact that the Implementation Year is only one year in the future, no existing analysis was performed. The Implementation Year (2020) analysis is considered sufficiently close to the existing conditions analysis for this Study. Existing traffic volumes (turning movement counts) were collected at the intersections targeted for analysis in this study in October 2019 and are included on Appendix Pages A-79 thru A-85. Adjustments were made for demand volumes.

Alameda Blvd. is a somewhat unique roadway corridor in two respects. First, it is a heavily travelled corridor connecting the east and west sides of the Albuquerque Metropolitan Area. There is a very heavy flow of traffic during the weekday AM Peak Hour period and a very heavy flow of traffic during the weekday PM Peak Hour period. Consequently, eastbound (AM) and westbound (PM) traffic back up on Alameda since the demand volumes exceed the system capacity. During the PM Peak Hour, westbound Alameda generally backs up from Corrales Rd. east to and beyond 2nd St. to Edith Blvd. The traffic volumes utilized in this study do consider the demand volumes at the intersections along Alameda Blvd. Drone videos were utilized to attempt to quantify the queuing at the two signalized intersections considered in this report.

Analysis of Implementation Year Conditions

Traffic Projections

Background traffic was taken from recent demand traffic counts (Appendix A-79 thru A-85) conducted for this project. A background traffic growth rate was applied to the existing traffic volumes to project the 2020 AM and PM Peak Hour volumes targeted for analysis.

This study assumes that the development will be implemented in one phase with an implementation year of 2020.

Projected trips were calculated based on the Institute of Traffic Engineers (ITE) Trip Generation Manual (10th Edition). Trips for the development were determined based on land use defined on

the Conceptual Site Development Plan on Page A-3 in the Appendix of this report. The following table summarized the trip generation rate for the project:

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Trip Generation Data (ITE Trip Generation Manual - 10th Edition)

USE (ITE CODE)		24 HR VOL	A. M. PEAK HR.		P. M. PEAK HR.	
DESCRIPTION		GROSS	ENTER	EXIT	ENTER	EXIT
Summary Sheet		Units				
Gasoline / Service Station w/ Convenience Market (945)	12	2,061	67	64	86	82
Subtotal		2,061	67	64	86	82
Pass-By Trips	30%		-20	-19	-26	-25
Total Primary Trips			47	45	60	57

A 30% pass-by trip rate was applied to this project. See Appendix Pages A-7 and A-8 for more information regarding the trip generation.

Primary and diverted linked trips for this commercial land use development were distributed proportionally to the 2020 projected population of Data Analysis Subzones within a two-mile radius of the proposed development. Population data for the years 2012 and 2040 were taken from the 2040 Socioeconomic Forecasts by Data Analysis Subzones for the MRCOG Region supplied by the Mid-Region Council of Governments (MRCOG). Population data from the years 2012 and 2040 was interpolated linearly to obtain 2020 population data to utilize for this analysis. Population Subzones were grouped based on the most likely major street(s) or route(s) to the subject development. The trip distribution worksheets and associated map data analysis subzones is shown on Appendix Pages A-9 thru A-11.:

The traffic signal system on Alameda is an adaptive system that optimizes itself in a real-time basis. Thus, there are no defined AM or PM Peak Hour signal timing plan that can be used for this analysis. Therefore, the signal timing was optimized using HCS7 software.

Background traffic growth rates were considered for each individual approach to an intersection that was targeted for analysis based on data from the 2008 through 2017 Traffic Flow maps prepared by the Mid-Region Council of Governments. Most of the Traffic Flow Data for those years taken from the MRCOG Traffic Flow Maps were Standard Data. The data from those years for each approach was plotted on a graph and a linear "regression trend line" calculated using the equation format $y=mx+b$. The growth rate was determined by calculating the average volume increase per year during the time period considered and dividing that volume into the most recent AWDT used in the analysis from which future volumes will be calculated. The rate of growth of that trend line was utilized as the annual growth rate for each approach if that calculated rate appeared feasible. However, in some roadway segments considered in this analysis, the rate indicated either an zero or a negative growth trend; therefore, the growth rate was considered to be a generic 0.5%. Historical Growth Rate Graphs with linear regression trend lines are shown in the Appendix on Pages A-16 thru A-22. The growth rate utilized for each approach to an

intersection is printed at the top of the Turning Movement sheets for each intersection (Appendix Pages A-24 thru A-25).

2020 PM Peak Hour BUILD Traffic Volumes

Traffic Analysis

A capacity analysis using optimized traffic signal timing was conducted for the Implementation Year (2020) NO BUILD and BUILD Conditions. The results are summarized as follows:

#1 – Alameda Blvd. / Fourth St. - Pages A-70 thru A-77

The results of the 2020 analyses of the signalized intersection of Alameda Blvd. / Fourth St. are summarized in the following table:

Alameda Blvd. / 4th St. 2020 Conditions	EB (Alameda Blvd.)			WB (Alameda Blvd.)			NB (4th St.)			SB (4th St.)		
	L	T	R	L	T	R	L	T	R	L	T	R
Existing Lane Geometry	1	2	1	1	2	1	1	1	1	1	1	1
AM Peak Hour												
2020 BUILD Conditions Volumes	164	2,468	104	15	556	15	120	52	35	91	68	116
V/C Ratio	0.25	1.00	0.09	0.18	0.26	0.02	0.50	0.28	0.23	0.41	0.40	0.81
Level-of-Service	A	D	A	C	A	A	D	D	D	D	D	E
Control Delay (Seconds)	5.2	35.4	6.1	32.9	5.9	5.4	47.5	50.4	50.1	47.3	52.1	61.8
Intersection LOS	C - 30.6											
95th Percentile Queue (ft.)	49.4	1,115	38.0	15.8	98.2	5.6	150.2	66.6	45.9	114.2	89.0	170.6

PM Peak Hour												
2020 BUILD Conditions Volumes	104	1,292	120	88	2,444	108	168	136	48	56	148	252
V/C Ratio	0.71	0.62	0.13	0.29	1.10	0.11	0.59	0.39	0.16	0.21	0.44	0.88
Level-of-Service	C	B	B	B	F	B	D	D	D	D	D	D
Control Delay (Seconds)	32.0	17.3	11.3	13.6	76.8	14.6	44.7	42.8	40.8	37.8	44.0	51.2
Intersection LOS	D - 50.5											
95th Percentile Queue (ft.)	82.8	422.2	66.2	37.3	1,348	91.6	86.5	161.6	54.4	60.6	179.0	305.1

The 2020 analysis of the intersection of Alameda Blvd. / Fourth St. demonstrates that the signalized intersection will operate at an acceptable level-of-service / delays with acceptable queuing except for the eastbound traffic during the AM Peak Hour and the westbound traffic during the PM Peak Hour. The purpose of this Study is not to measure the impact of the proposed Murphy Express on the existing signalized intersection. It is well known that there is congestion and excessive queuing at signalized intersections along Alameda Blvd. during the AM and PM Peak Hour periods. However, analysis of the signalized intersections to the east and west of the proposed Murphy Express driveway are necessary to measure the operation of the proposed new driveway.

#2 – Alameda Blvd. / Second St. - Pages A-70 thru A-77

The results of the 2020 analyses of the full access signalized intersection of Alameda Blvd. / Second St. are summarized in the following tables:

Alameda Blvd. / 2nd St. 2020 Conditions	EB (Alameda Blvd.)			WB (Alameda Blvd.)			NB (2nd St.)			SB (2nd St.)		
	L	T	R	L	T	R	L	T	R	L	T	R
Existing Lane Geometry	1	2	1	1	2	1	2	2>	0	1	2>	0
AM Peak Hour												
2020 BUILD Conditions Volumes	64	1,596	756	16	448	24	160	156	216	40	112	40
V/C Ratio	0.10	0.73	0.77	0.07	0.20	0.02	0.29	0.53	0.86	0.30	0.30	0.32
Level-of-Service	A	A	A	A	A	A	D	D	D	D	D	D
Control Delay (Seconds)	6.8	4.6	4.0	9.2	9.6	8.4	41.1	47.2	52.8	43.3	46.8	46.9
Intersection LOS	B - 13.3											
95th Percentile Queue (ft.)	25.2	123.9	80.6	6.3	109.2	10.7	89.9	192.5	266.5	45.6	93.5	91.3

PM Peak Hour												
2020 BUILD Conditions Volumes	68	624	472	400	1,984	64	408	188	76	44	252	84
V/C Ratio	0.54	0.42	0.71	0.69	0.96	0.07	0.70	0.36	0.38	0.17	0.78	0.81
Level-of-Service	C	B	B	B	D	B	D	D	D	D	D	D
Control Delay (Seconds)	29.9	11.2	14.3	12.9	35.7	11.1	40.4	41.6	41.9	43.9	53.6	54.5
Intersection LOS	C - 29.3											
95th Percentile Queue (ft.)	55.3	163.5	199.5	210.9	923.2	34.6	225.7	158.2	151.4	51.8	224.7	217.5

The 2020 analysis of the intersection of Alameda Blvd. / Second St. demonstrates that the signalized intersection will operate at an acceptable level-of-service / delays with acceptable queuing except for the eastbound traffic during the AM Peak Hour and the westbound traffic during the PM Peak Hour. The purpose of this Study is not to measure the impact of the proposed Murphy Express on the existing signalized intersection. It is well known that there is congestion and excessive queuing at signalized intersections along Alameda Blvd. during the AM and PM Peak Hour periods. However, analysis of the signalized intersections to the east and west of the proposed Murphy Express driveway are necessary to measure the operation of the proposed new driveway.

Overall, the analyses of the signalized intersections of Alameda Blvd. / Second St. do not seem to quite reflect the degree of congestion along the corridor during the peak hour periods. The 95th Percentile Queues seem to be under-reported when compared to actual field conditions, especially the eastbound thru movements on Alameda Blvd. at 4th St. and the westbound thru movements on Alameda Blvd. at 2nd St. Actual field delays are difficult if not impossible to determine since the queuing is so large.

The current geometry of the southbound approach of 2nd St. at Alameda Blvd. is a single left turn lane, a thru lane, and a thru / right turn lane. The County has questioned whether or not a right turn lane would be warranted based on the New Mexico Department of Transportation's State

Access Management Manual. Chapter 7 (Page 80) of the NM DOT's State Access Management Manual states:

The use of speed-change lanes at signalized intersections is generally consistent for all access categories, urban and rural. Guidelines for determining the need for speed-change lanes at signalized intersections are provided below. The guidelines apply to all access categories except UINT and RINT. Situations where guidelines vary by access category are noted.

(a) Exclusive Right-turn Lanes: Exclusive right-turn lanes should be considered at signalized intersections under the following conditions:

- i. Where the right-turn design hour volume (DHV) equals or exceeds 300 DHV and the volume in the outside general purpose travel lane equals or exceeds 300 DHV (i.e., the total volume in the outside travel lane is equal to or greater than 600 DHV including a right-turn volume of at least 300 DHV); or,
- ii. Where the right-turn volume equals or exceeds 150 DHV and the volume-to-capacity (v/c) ratio for the adjacent through movement(s) is expected to be 0.85 or greater based on accepted analysis methodologies; or,
- iii. Where the right-turn volume equals or exceeds 100 DHV and the posted speed is 45 mph or above.

The posted speed limit on 2nd St. north of Alameda is 45 MPH, so criteria iii would apply (where the right turn volume equals or exceeds 100 DHV and the posted speed is 45 mph or above). The forecast southbound right turn volumes on 2nd St. at Alameda Blvd. is 40 vph during the 2020 AM Peak Hour and 84 vph during the 2020 PM Peak Hour. Therefore, a southbound right turn lane is not warranted at the signalized intersection of Alameda Blvd. / 2nd St.

#3 –Alameda Blvd. / Driveway “A” – Pages A-78 thru A-79

The results of the analysis of the unsignalized intersection of Alameda Blvd. / Driveway “A” are summarized in the following table:

Alameda Blvd. / Driveway "A" 2020 Conditions	EB (Alameda Blvd.)			WB (Alameda Blvd.)			NB (Driveway "A")			SB (Driveway "A")		
	L	T	R	L	T	R	L	T	R	L	T	R
Existing Lane Geometry	1	2			2>	0				1		1
AM Peak Hour												
2020 BUILD Conditions Volumes	29	2,245			775	38				45		19
V/C Ratio	0.03									0.34		0.03
Level-of-Service	A									E		A
Control Delay (Seconds)	8.2									46.2		9.8
Intersection LOS	TWSC											
95th Percentile Queue (veh)	0.1									1.4		0.1
With Driveway "B"		2			2>	0						1
2020 BUILD Conditions Volumes		2,245			775	38						19
V/C Ratio												0.03
Level-of-Service												A
Control Delay (Seconds)												9.8
Intersection LOS	TWSC											
95th Percentile Queue (veh)												0.1

PM Peak Hour												
2020 BUILD Conditions Volumes	26	1,158			2,458	60				48		35
V/C Ratio	0.30									1.46		0.60
Level-of-Service	F									F		F
Control Delay (Seconds)	64.1									503.0		135.0
Intersection LOS	TWSC											
95th Percentile Queue (veh)	1.1									5.3		2.5
With Driveway "B"		2			2>	0						1
2020 BUILD Conditions Volumes		1,158			2,458	60						35
V/C Ratio												0.60
Level-of-Service												F
Control Delay (Seconds)												135.4
Intersection LOS	TWSC											
95th Percentile Queue (veh)												2.5

The 2020 analysis of the intersection of Alameda Blvd. / Driveway “A” serving the Murphy Express demonstrates that the operation will be marginally acceptable during the AM Peak Hour period, but the southbound exiting traffic from the driveway will experience long delays during the PM Peak Hour period. This is probably the case for most all of the existing driveways along this section of Alameda Blvd. Many of the existing driveways are full access. What occurs at these driveways during the peak hour period is that congestion on Alameda Blvd. results in traffic

slowing to less than about 15 MPH or less, so Alameda drivers stop to let driveway traffic negotiate the turn into or out of the driveways. Even though the preceding table reports that the eastbound left turn movement into Driveway "A" and the southbound right turn exiting traffic both have long delays, a SimTraffic microsimulation of the driveway shows that both movements operate pretty well with reasonable delays. The microsimulation shows that the southbound left turn movement exiting Driveway "A" to travel eastbound on Alameda will experience long delays with long queuing. This report recommends that the eastbound left turn into Driveway "A", the westbound right turn into Driveway "A", and the southbound right turn into Driveway "A" be approved. The approval of the southbound left turn exiting Driveway "A" to travel eastbound on Alameda will be determined by the New Mexico Department of Transportation. In consideration of the numerous full access driveways along Alameda Blvd., the conditions seem to allow for full access operation of such driveways under highly congested and impeded traffic flow. Due to safety concerns, though, it is unlikely that the New Mexico Department of Transportation will approve the southbound left turn movement exiting Driveway "A" to travel eastbound on Alameda Blvd. It should be noted that restricting this left turn will likely generate U-turns on Alameda to the west of Driveway "A" unless an alternative access to Second St. is provided. This study finds that the alternative access to Second St. is preferable in that it will help remedy the operational and safety issues related to the problematic southbound left turn exiting movement at Driveway "A". The projected operation of the alternative access onto Second St. (Driveway "B") is summarized in the following table:

Driveway "B" / Second St. 2020 Conditions	EB (Driveway "B")			NB (Second St.)			SB (Second St.)		
	L	T	R	L	T	R	L	T	R
Existing Lane Geometry	1>				<2			2>	
AM Peak Hour									
2020 BUILD Conditions Volumes	6		39	5	292			340	7
V/C Ratio									
Level-of-Service	A			A					
Control Delay (Seconds)	9.9			8.1					
Intersection LOS									
TWSC									
95th Percentile Queue (veh)	0.2			0.0					
PM Peak Hour									
2020 BUILD Conditions Volumes	7		41	5	320			380	6
V/C Ratio	0.07			0.01					
Level-of-Service	B			A					
Control Delay (Seconds)	10.1			8.2					
Intersection LOS									
TWSC									
95th Percentile Queue (veh)	0.2			0.0					

Driveway "B" will operate satisfactorily, especially when compared to the operation of the southbound left turn movement were it to occur at Driveway "A".

In consideration that the NM DOT has mandated that Driveway "A" will be restricted to a right-in, right-out only driveway the analysis of Driveway "A" in the table above shows the results based on the restriction imposed by the NM DOT. The southbound right turn out is still at LOS "F". This analysis assumes that the loss of the ability to make the left turn in (eastbound) at Driveway "A" will impact the trip generation rate adversely. Thus, the right-in and right-out volumes would be virtually

the same as if the driveway were full access. With the southbound left-turn-out movement being prohibited, the exiting traffic desiring to travel eastbound on Alameda Blvd. or southbound on Second St. will opt to exit at Driveway “B” on Second St.

The Bernalillo County Public Works Department has commented that Driveway “B” on 2nd St. should be restricted to a right-in, right-out driveway.

Access Design Specifications

Sight distance at Driveway “A” is adequate. There are no vertical or horizontal curves along this portion of Alameda Blvd. (NM S.R. 528) and there are no structures that are blocking sight distance into and out of the driveway.

Sight distance at Driveway “B” is adequate. There are no vertical or horizontal curves along this portion of Second St. and there are no structures that are blocking sight distance into and out of the driveway. There is a horizontal curve on Second St. to the north of the proposed driveway, but sight distance is easily at least 500 feet which is adequate for a driveway on a 45 MPH roadway.

The Futures 2040 Metropolitan Transportation Plan (2040 Long Range Bikeway System Map) depicts Alameda Blvd. as having a bicycle lane and a paved trail along this segment of the roadway. Therefore, any design and construction of a right turn deceleration lane will need to accommodate the bicycle lane on the north side of Alameda Blvd.

Based on Table 17.B-2 (Criteria for Deceleration Lanes on Urban Multi-Lane Highways) in the New Mexico Department of Transportation’s State Access Management Manual, both a westbound right turn deceleration lane and an eastbound left turn deceleration lane are warranted. The required lengths of the right and left turn lanes (based on a 40 MPH posted speed limit) are 300 feet long including transition and 325 feet long including transition respectively. Due to the fact that this project is located in a developed urban area, there are physical constraints that preclude the auxiliary lanes from being designed and constructed to the required lengths. This report recommends that the westbound right turn deceleration lane be constructed to the maximum length possible (approximately 200 feet long) between the west end of the curb return at Alameda / 2nd St. and the new Driveway “A”. It also recommends that the existing center two-way left turn lane be utilized as an eastbound left turn lane into the new Driveway “A”. The calculated 95th Percentile queue length for the eastbound left turn movement into Driveway “A” is one vehicle – see Page A-79 in the Appendix.

Bernalillo County Transportation Planner, Julie Luna, has commented that “Recommendation for full access on 2nd St at northern end of parcel - This full access is within the functional area of the intersection of Alameda Blvd. and 2nd St. It does not meet Bernalillo County’s required 250 ft. separation from face of curb to driveway centerline. A variance is required for the proposed access on 2nd St. Previous correspondence outlined concerns with the proposed 2nd St. access.

Unless a variance is approved, the recommendation for access onto 2nd St. within the TIS needs to be removed.” Further discussions with the Bernalillo County Public Works Department and the New Mexico Department of Transportation indicate that the proposed driveway on 2nd St. (Driveway “B”) can be approved as a right-in, right-out only driveway located approximately 200 feet north of Alameda Blvd. (centerline to centerline).

Additionally, the New Mexico Department of Transportation has commented that “Driveway “A” (on Alameda Blvd.) shall be limited to a partial right-in / out access.”

Therefore, this Study acknowledges that the Bernalillo County Public Works Department with concurrence by the New Mexico Department of Transportation will restrict access at Driveway “B” on 2nd St. to a right-in, right-out only unsignalized driveway, and that Driveway “A” on Alameda Blvd. will be restricted to right-in, right-out driveway.

The constrained access mandated by the New Mexico Department of Transportation and the Bernalillo County Public Works Department will make it difficult for certain customers of Murphy Express to access the facility from certain directions and to exit the facility to travel certain directions. Traffic desiring to travel to the Murphy Express travelling eastbound on Alameda Blvd. from the west of the facility will be required to negotiate a circuitous route to access the new Murphy Express facility. The circuitous route could involve routing in a roundabout manner and sometimes include U-Turns at some location. Traffic exiting the Murphy Express facility desiring to travel northbound on 2nd St. will also need to take a circuitous route to achieve their desired direction of travel. Fortunately, the percentage of traffic desiring the travel northbound on 2nd St. is expected to be minor.

Summary of Deficiencies, Anticipated Impacts, and Recommendations

The implementation year 2020 analysis did not determine any significant deficiencies in the adjacent transportation system other than to confirm the congestion and associated long queues on Alameda Blvd. during the peak hour periods. This has been a known phenomenon for the past 20 years or more. The LOS / Volumes / Geometry map at the end of this report summarize the 2020 BUILD Conditions analyzed for the two signalized intersection as well as the proposed Driveway "A" in this Study. This Study finds the proposed Driveway "A" to be reasonably acceptable when considering the eastbound left turn into the driveway, the westbound right turn into the driveway, and the southbound right turn exiting the driveway. Permitting the southbound left turn exiting the driveway to travel westbound onto Alameda Blvd. will probably result in excessive delays and long queuing for the southbound left turn traffic. There is not much room within the Murphy site to contain such queuing. Prohibiting the southbound left turn movement will help solve the issues associated with the left turns, but will likely result in rerouted traffic that will seek alternate routes to their destination or will execute U-turns downstream on Alameda Blvd. to the west of Driveway "A" so as to travel eastbound. This report recommends additional access onto Second St. north of Alameda Blvd.

Recommendations:

Based on 2020 (Implementation Year) Analysis:

- An eastbound left turn deceleration lane and a westbound right turn deceleration lane are both warranted at the proposed Driveway "A" accessing the new Murphy Express facility. Based on the posted speed limit of 40 MPH on Alameda Blvd. along the frontage of this project, the New Mexico Department of Transportation's State Access Management Manual stipulates that the right turn deceleration lane should be designed and constructed to a length of 300 feet including transition. It stipulates that the left turn deceleration lane should be designed and constructed to a length of 325 feet including transition. Field constraints imposed by urban conditions preclude construction of the right turn and left turn deceleration lanes warranted. This study recommends that the westbound right turn deceleration lane be designed and constructed to begin at the west curb return of the intersection of Alameda Blvd. / 2nd St. and extend about 200 feet to the new Murphy Express Driveway "A". By virtue of the fact that the New Mexico Department of Transportation has prohibited the eastbound left turn movement into Driveway "A" off of Alameda Blvd., then warrant for the eastbound left turn lane is rendered meaningless.
- Provide a new connection to and from the proposed Murphy Express facility to a new driveway on Second St. which will be restricted to right-in, right-out only by the Bernalillo County Public Works Department. The new driveway on Second St. (Driveway "B") should be located as far as possible north of Second St., but a minimum of 250 feet (centerline to centerline) if possible. Access from the Murphy Express facility to Second St. will require coordination and permitting from the Middle Rio Grande Conservancy District (MRCGD) to cross the existing drainage ditch along the west side of Second St.

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APPENDIX

Attachment: BA2026-0001_Additional Documents (26992 : ZBA2026-0001/ZA2026-0024)